Understanding Relationship Between Variables

Business Scenario:
A company increases its advertising budget. Will it impact on sales increament?

If advertising budget increases, will sales always increase?
How can we check the relationship using the data?
Can two variables move together without causing the other?

A company records its advertising expenditure and sales.

Month	Spend(th)	Sales(th)
1	20	30
2	40	50
3	60	70
4	80	90
5	100	110

What happens to sales when advertising spending increases?

Marketing teams analyze advertising expenditure and sales to understand whether higher spending associated with higher revenue.

Covariance

It tells us whether two variables tend to move together or in opposite directions.

Techincal:
Sample covariance measures the joint varitaion of two variables around their respective sample means.

$$Cov(X,Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Xi,Yi: Individual observations
Xbar, Ybar = sample means
n: Number of observations.

Types:
Positive: Variables tend to move in the same direction
Ex: Advertising spend and sales
Negative : Variables tend to move in opposite directions
Ex: Product price and demand
Near zero : Little linear co -movement
Ex: shoe size and exam score.

Practical calcualation:

	X	Y	X-Xbar	Y-Ybar	Product
	20	30	-40	-40	1600
	40	50	-20	-20	400
	60	70	0	0	0
	80	90	20	20	400
	100	110	40	40	1600
Average	60	70			

Cov(X,Y) =(1600+400+0+400+1600)/5-1

Cov(X,Y)=	1000	1000

The positive result indicates that both variables move in the same direction.

Limitation: Covariance depends on the units of measurement, so its magnitude is difficult to compare across datasets.

Correlation

Correlation tells us the strength and direction of the relationship between two variables.

Techincally:
It is a standardized measure of association between variables. Pearson correlation measures the strength and direction of their relationship.

	Covariance	**Correlation**
Purpose	Direction of joint variation	Direction and strength of linear relationship
Range	No fixed range	-1 to +1
Units	Depends on measurement units	Unit free
Interpret	magnitude can be difficult to compare	Easier to interpret
Excel	COVARIANCE.S()	CORREL()

Business Examples:

1. Advertising and sales
Advertising spend increases, and sales generally increase.
+ve correlation

2. Price and Demand
Product price increase, and demand may decrease
-ve correaltion

3. Shoe Size and employee productivity
There is no meaningful linear relationship.
correlation near 0

If two variables have strong correlation, does that prove one can cause other?
N. Correlation alone does not establish causation.

Pearson Correlation Coefficient (r)

Pearson's correlation coeficient measures the strength and the direction of the linear relationship between two numerical variables.

Formula:

$$r = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}}$$

Properties:

$$-1 \leq r \leq +1$$

r = +1: Perfect postive linear relationship
r = -1: Perfect negative linear relationship
r = 0 : No linear relationship

X	Y	y = X+10
20	30	30
40	50	50
60	70	70
80	90	90
100	110	110

The relationship is perfectly linear because:

$$Y = X + 10$$

$$r = +1$$

These five obseravtions show a perfect +ve linear relationship. This is deliberately clean example to understand +ve perfect correalation, although it's rare to get r = +1 in real data.

Suppose the correlation between advertising spend and sales is r = +0.82

Interpretaion:

The relationship is positive
The linear association is strong
Higher advertising spend tends to be associated with higher sales
This does not prove advertising caused the increase.

0.9	Strong +ve
0.45	Moderate +ve
-0.78	Strong -ve
-0.15	Weak -ve
0	No linear association

Range and Interpretaion of r

Range of r	General Interpretaion
+0.70 to +1.00	Strong Positive
+0.30 to +0.69	Moderate Positive
+0.01 to 0.29	Weak Positive
0	No linear correlation
- 0.01 to -0.29	Weak Negative
-0.30 to -0.69	Moderate Negative
-0.70 to -1.00	Strong Negative

Case 1: Retail

Correlation between discount percentage and sales valume:

r = +0.68

Higher discounts are associated with higher sales volume.

Business Q: Is the additional sales volume enough to compensate for the reduced profit margin?

Case 2 : HR Analytics

Correlation between training hours and employee performance:

r = +0.68

Employees with more training hours tend to have higher performance scores.

Question: Does the relationship remain after accounting for experience, role and prior performance?

Case 3: E-Commerce

Correlation between delivery time and customer satisfaction:

r = -0.72

Longer delivery times tend to be associated with lower customer satisfaction

Question: Which delivery stages contribute most to the delays?

Correlation vs Causation

Correlation:

Two variables are associated or tends to change together.

Causation:

A change in one variable produces a change in another under the relavant casual conditions.

Example 1: Ice cream sales and drowning incidents

Both may increase during the summer.

Does ice cream cunsumption cause drowning incidents?

No. Hot weather and increased swimming activity may help exaplain why both increase.

This is an example of **confounding variable**

Example 2: Advertising and sales

Sales and advertising expenditure may be positively correlated.

However:

A company may increase advertising during peak season.

Product demand may already be increasing

Discounts and competitor activity may affect sales.

Correlation alone cannot isolate the casual effect of advertising.

Example 3: Training hours and employee performance

Employees who receive more training may perform better.

But experienced employees may receive more training, or high performing emloyees may be selected for advanced training.

The relationship needs further investiagation

How to investigate causation

Conduct a randomized experiment when practical

Use an appropriate control group

Consider confounding variables

Use suitable casual inference methods when experiments are not possible

<u>Scattor Plots</u>

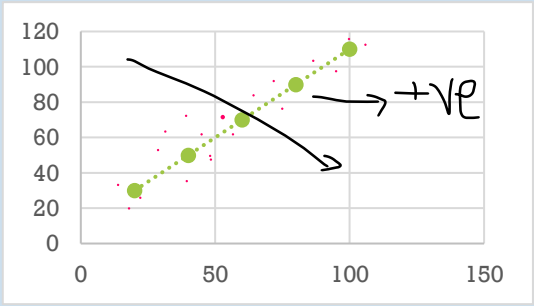
A scatter plot displays paired numerical observations as points on a grapgh to help visualize relationship

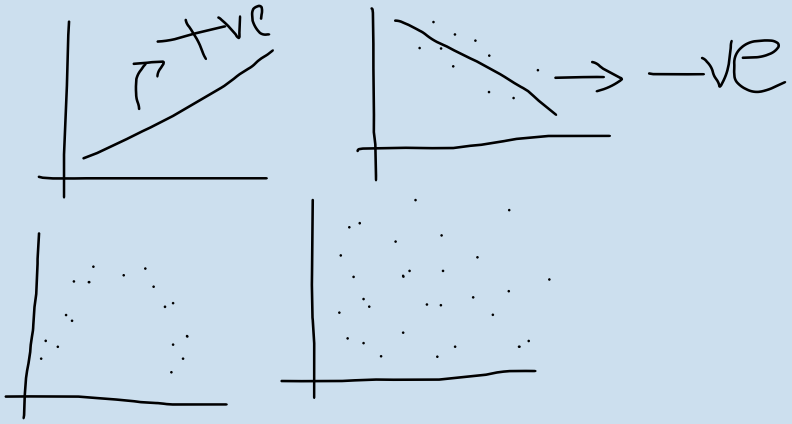
X- axis: Explainatory or independent variable

Y- axis : Response or dependent variable

Each point : one paired observation

X	Y
20	30
40	50
60	70
80	90
100	110





A retail analyst can plot discount percentage against units sold to examine whther higher discounts tend to be associated with higher sales

Campaign	Spend	Sales
A	10	20
B	20	25
C	30	40
D	40	42
E	50	60
Covarianc	242.5	0.9723364

